

Network motifs

MCSB Bootcamp — Mathematical and Computational Track

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```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp
```

1. Direct negative feedback

Gene A's protein represses gene A. *Kaa* is the strength of that repression, and *gamma_ma0* the baseline rate of mRNA production it acts on.

The three parameter sets are chosen so that the *steady state* is roughly the same in all three; what changes is how the system gets there.

```
delta_ma = 0.05
gamma_pa = 0.02
delta_pa = 0.01

params = [
    {"Kaa": 0, "gamma_ma0": 1},
    {"Kaa": 0.1, "gamma_ma0": 5},
    {"Kaa": 1, "gamma_ma0": 40},
]

fig, axes = plt.subplots(1, len(params), figsize=(11, 4))

for i_param, ax in zip(range(len(params)), axes):

    Kaa = params[i_param]["Kaa"] # Strength of inhibition of A by A
    gamma_ma0 = params[i_param]["gamma_ma0"] # Baseline production rate mRNA for A

    def dxdt(t, state):
```

```

"""dma/dt and dpa/dt for a gene whose protein represses it."""
ma, pa = state

gamma_ma = gamma_ma0 / (1 + Kaa * pa)

dma_dt = +gamma_ma - delta_ma * ma
dpa_dt = +gamma_pa * ma - delta_pa * pa

return [dma_dt, dpa_dt]

sol = solve_ivp(dxdt, [0.0, 600], [0, 0])

T = sol.t
X = sol.y.T

ax.plot(T, X[:, 0], "-g", linewidth=3) # green for RNA
ax.plot(T, X[:, 1], "-", color=[0.5, 0, 1], linewidth=3) # purple for protein
ax.set_xlim(0, 600)
ax.set_ylim(0, 80)
ax.set_box_aspect(1)
ax.set_ylabel("RNA and Protein")
ax.set_xlabel("Time (seconds)")

fig.tight_layout()
plt.show()

```

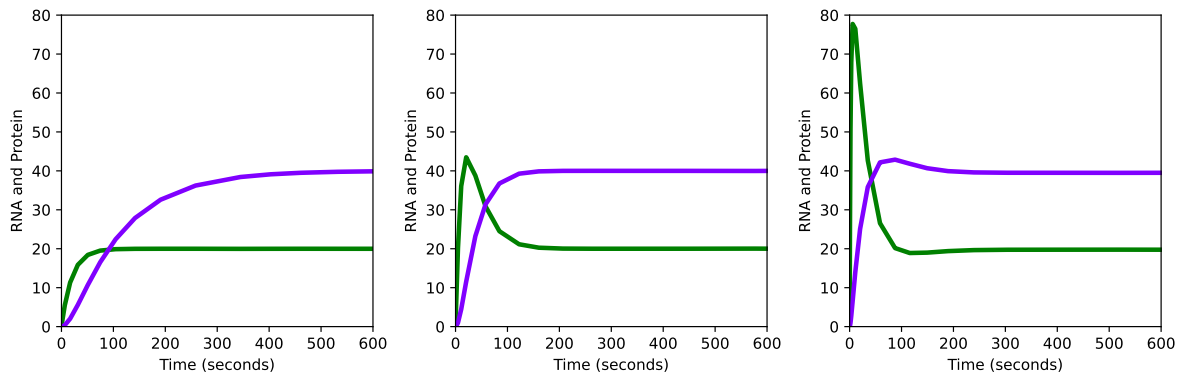


Figure 1: Direct negative feedback at three strengths, from none on the left to strong on the right.

2. Indirect negative feedback

Gene A activates gene B, and gene B's protein represses gene A. An external input signal drives A's production, stepping up at $t = 60$ s and again at $t = 2000$ s.

```
delta_ma = 0.08
gamma_pa = 1
delta_pa = 1

delta_mb = 1e-4
gamma_pb = 1
delta_pb = 1

params = [
    {"title": "No feedback", "Kba": 0, "Kab": 1e4, "t_max": 400},
    {"title": "Feedback", "Kba": 1.0, "Kab": 1e4, "t_max": 400},
    {"title": "Feedback, long timescale", "Kba": 1.0, "Kab": 1e4, "t_max": 4000},
]

# an input signal, e.g., externally-induced production
def input_signal(t):
    return 0 * (t < 60) + 100 * (t > 60) + 300 * (t > 2000)

for i_param in range(len(params)):

    Kba = params[i_param]["Kba"] # Strength ( $IC_{50}^{-1}$ ) of inhibition of A by B
    Kab = params[i_param]["Kab"] # Strength ( $EC_{50}^{-1}$ ) of activation of B by A
    t_max = params[i_param]["t_max"]

    def dxdt(t, state):
        """A driven by an input and repressed by B, B activated by A."""
        ma, pa, mb, pb = state

        gamma_ma = 1 / (1 + Kba * pb) * (1.0 + input_signal(t))
        gamma_mb = 1e-3 * (1 + Kab * pa)

        dma_dt = +gamma_ma - delta_ma * ma
        dpa_dt = +gamma_pa * ma - delta_pa * pa

        dmb_dt = +gamma_mb - delta_mb * mb
        dpb_dt = +gamma_pb * mb - delta_pb * pb
```

```

    return [dma_dt, dpa_dt, dmb_dt, dpb_dt]

# Start well before t = 0 so the model has settled by the time the input arrives.
sol = solve_ivp(dxdt, [-10000.0, t_max], [20, 20, 0, 0])

# Matlab's y-axis autoscales to the data inside the current x-limits; matplotlib's autos
# The run starts at t = -10000 only so that the model has settled by the time the input a
visible = sol.t >= 0
T = sol.t[visible]
X = sol.y.T[visible]

fig, (ax_input, ax_a, ax_b) = plt.subplots(3, 1, figsize=(6, 7))
ax_input.set_title(params[i_param]["title"])

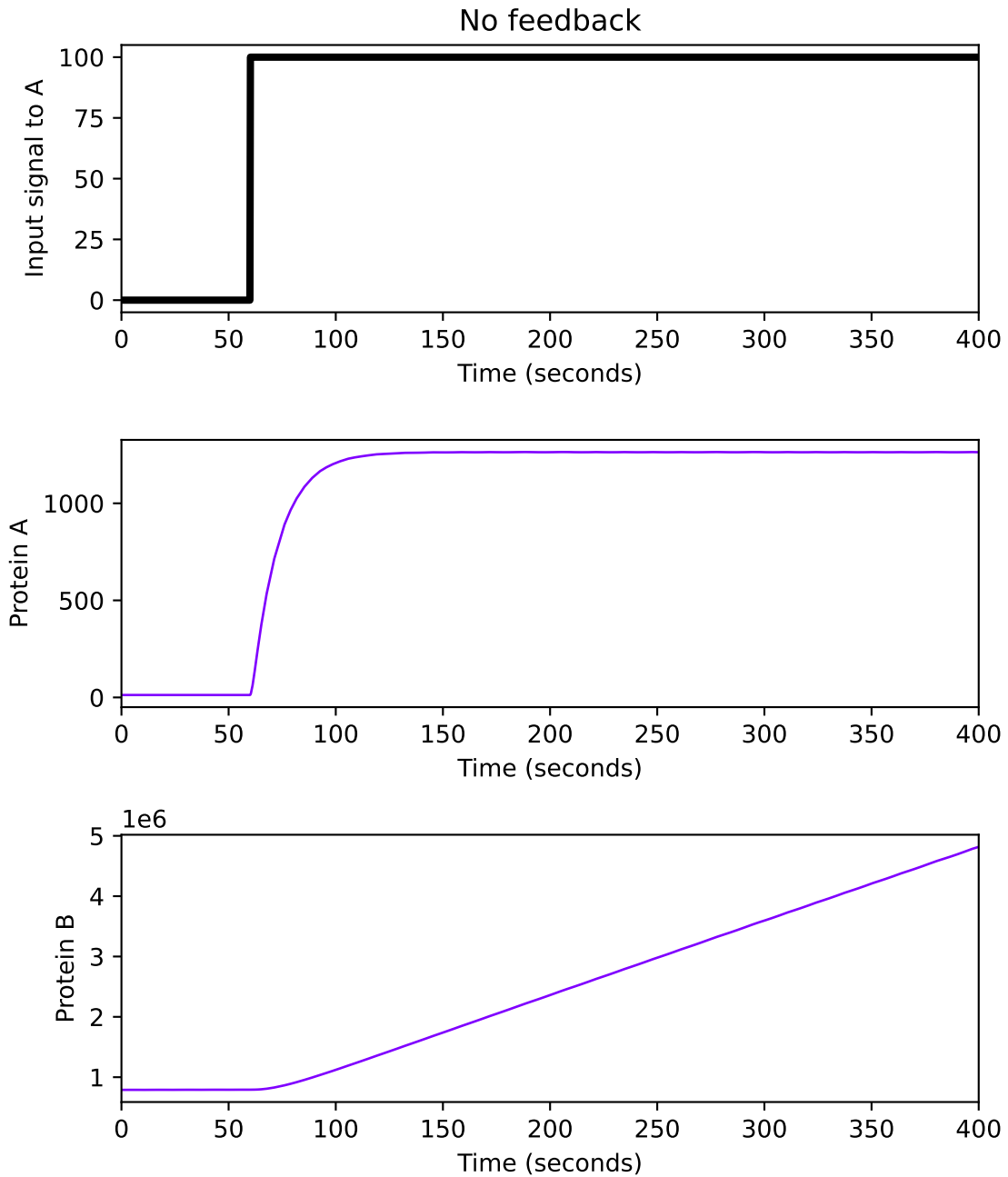
ax_input.plot(T, input_signal(T), "-k", linewidth=3)
ax_input.set_ylabel("Input signal to A")
ax_input.set_xlabel("Time (seconds)")
ax_input.set_xlim(0, t_max)

ax_a.plot(T, X[:, 1], "-", color=[0.5, 0, 1], linewidth=1) # purple
ax_a.set_ylabel("Protein A")
ax_a.set_xlabel("Time (seconds)")
ax_a.set_xlim(0, t_max)

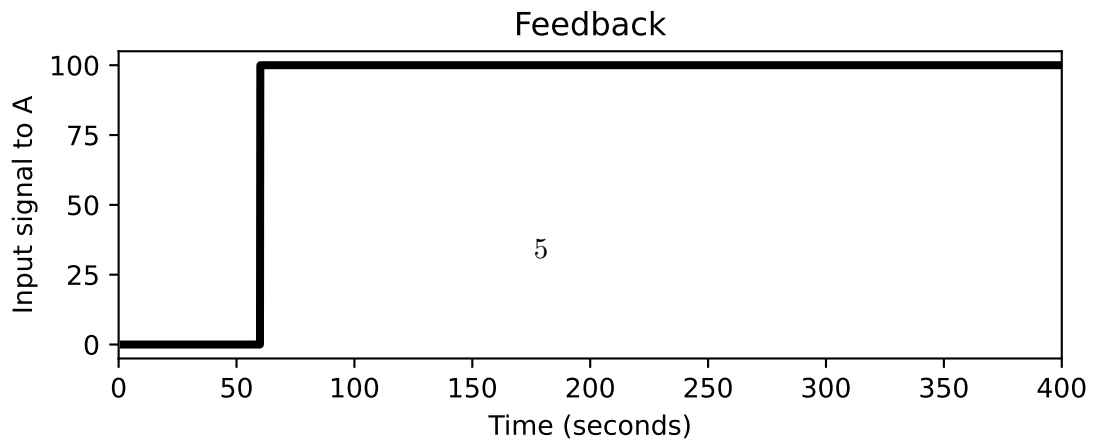
ax_b.plot(T, X[:, 3], "-", color=[0.5, 0, 1], linewidth=1) # purple for protein
ax_b.set_ylabel("Protein B")
ax_b.set_xlabel("Time (seconds)")
ax_b.set_xlim(0, t_max)

fig.tight_layout()
plt.show()

```



(a) Indirect negative feedback: the same step input, without feedback, with feedback, and with feedback watched for long enough to see the second step.



3. Double negative feedback

A represses B and B represses A, with every parameter jittered a little from run to run. This is the motif that makes a switch: which of the two genes wins is decided by where the run happens to start.

```
rng = np.random.default_rng(6)

noisiness = 0.1
noisiness2 = 0.1

num_runs = 1000
storage_A = np.zeros(num_runs)
storage_B = np.zeros(num_runs)

fig, (ax_a, ax_b) = plt.subplots(2, 1, figsize=(6, 7))

for i_run in range(num_runs):

    delta_ma = 1 * (1 + noisiness * rng.random())
    gamma_pa = 1 * (1 + noisiness * rng.random())
    delta_pa = 1 * (1 + noisiness * rng.random())

    delta_mb = 1 * (1 + noisiness * rng.standard_normal())
    gamma_pb = 1 * (1 + noisiness * rng.random())
    delta_pb = 1 * (1 + noisiness * rng.random())

    Kba = 1 * (1 + noisiness2 * rng.standard_normal()) # Strength (IC50-1) of inhibition of B by A
    Kab = 1 * (1 + noisiness2 * rng.standard_normal()) # Strength (IC50) of inhibition of B by A

    def dxdt(t, state):
        """A and B, each repressing the other."""
        ma, pa, mb, pb = state

        gamma_ma = 3 / (1 + abs(Kba * pb) ** 3)
        gamma_mb = 3 / (1 + abs(Kab * pa) ** 3)

        dma_dt = +gamma_ma - delta_ma * ma
        dpa_dt = +gamma_pa * ma - delta_pa * pa

        dmb_dt = +gamma_mb - delta_mb * mb
        dpb_dt = +gamma_pb * mb - delta_pb * pb
```

```

        return [dma_dt, dpa_dt, dmb_dt, dpb_dt]

initial_condition = 2 * rng.random(4)

sol = solve_ivp(dxdt, [0.0, 60], initial_condition)

T = sol.t
X = sol.y.T

if i_run < 9:
    ax_a.plot(T, X[:, 0], "-r") # red for RNA
    ax_a.plot(T, X[:, 1], "-", color=[0.5, 0, 1]) # purple
    ax_a.set_ylabel("RNA and Protein A")
    ax_a.set_xlabel("Time (min)")

    ax_b.plot(T, X[:, 2], "-r") # red for RNA
    ax_b.plot(T, X[:, 3], "-", color=[0.5, 0, 1]) # purple
    ax_b.set_ylabel("RNA and Protein B")
    ax_b.set_xlabel("Time (min)")

storage_A[i_run] = X[-1, 1]
storage_B[i_run] = X[-1, 3]

fig.tight_layout()
plt.show()

```

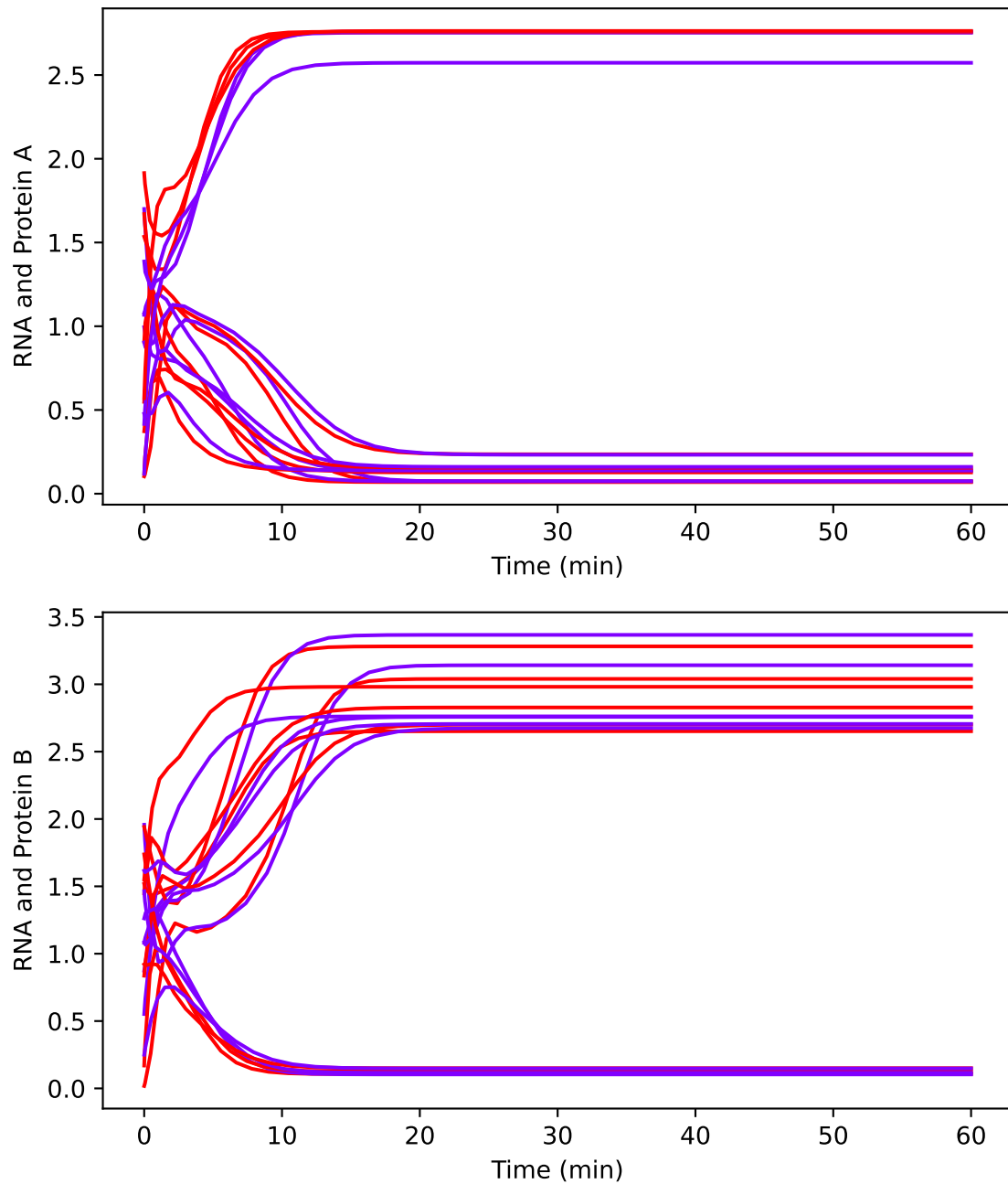


Figure 3: Nine runs of the double-negative-feedback motif from random starting points.

```
fig, (ax_a, ax_b) = plt.subplots(1, 2, figsize=(9, 4.5))
ax_a.hist(storage_A, 50)
```



```

ax_a.set_box_aspect(1)
ax_a.set_xlabel("Amount of Protein A")

ax_b.hist(storage_B, 50)
ax_b.set_box_aspect(1)
ax_b.set_xlabel("Amount of Protein B")

fig.tight_layout()
plt.show()

```

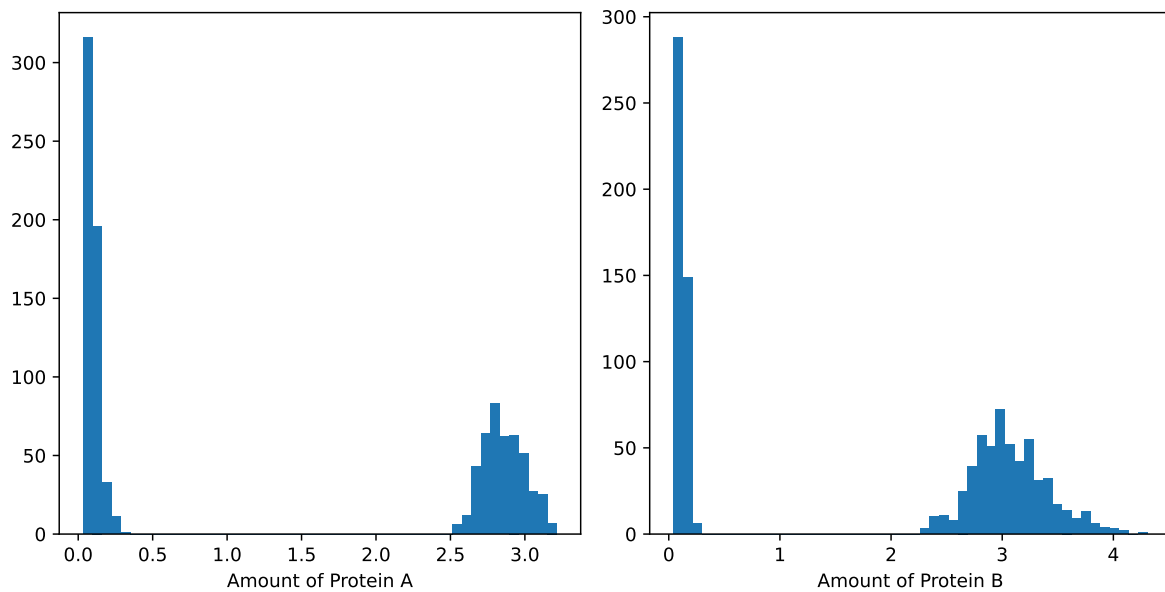


Figure 4: Where the thousand runs ended up, one histogram per gene.

```

fig, ax = plt.subplots(figsize=(5, 5))
ax.scatter(storage_A, storage_B, 5)
ax.set_box_aspect(1)
ax.set_xlabel("Amount of Protein A")
ax.set_ylabel("Amount of Protein B")
plt.show()

```

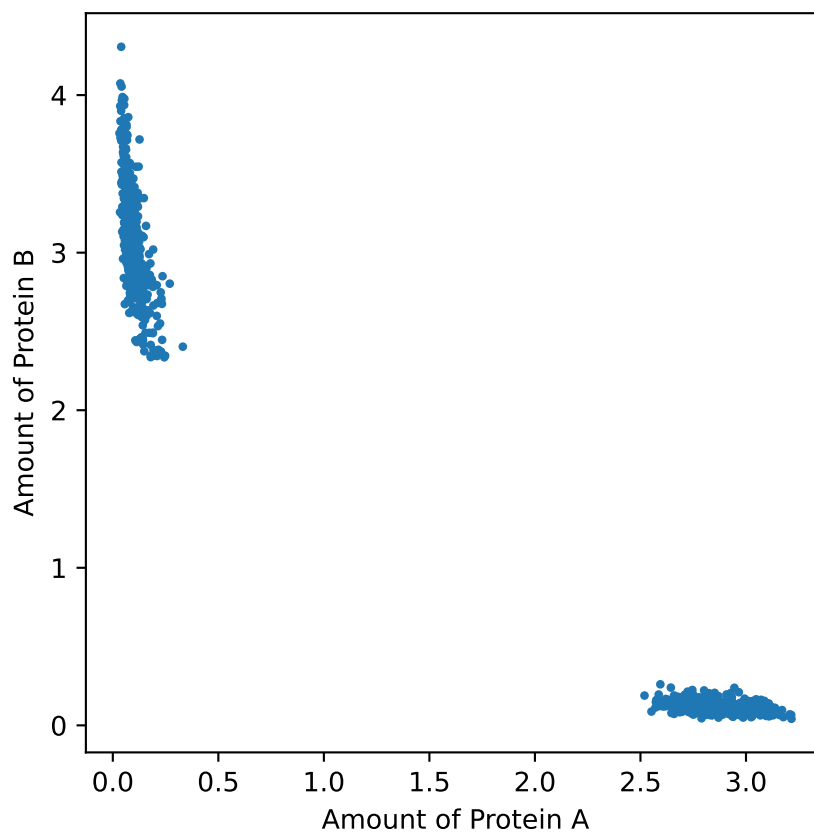


Figure 5: The same thousand runs, plotted against each other.

4. Incoherent feed-forward

A activates B, A activates C, and B represses C. C therefore receives the same signal twice, once with each sign, and by two paths of different lengths.

```
delta_ma = 0.05
gamma_pa = 0.1
delta_pa = 0.02

delta_mb = 0.05
gamma_pb = 0.1
delta_pb = 0.02

delta_mc = 0.05
gamma_pc = 0.1
```

```

delta_pc = 0.02

gamma_ma = 10

Kab = 10 # Strength (EC50) of activation of B by A
Kac = 10
Kbc = 10 # Strength (IC50) of inhibition of C by B

def dxdt(t, state):
    """A drives B and C; B represses C."""
    ma, pa, mb, pb, mc, pc = state

    gamma_mb = 10 * (1 + (Kab * pa))
    gamma_mc = 10 * (1 + (Kac * pa)) / (1 + Kbc * pb)

    dma_dt = +gamma_ma - delta_ma * ma
    dpa_dt = +gamma_pa * ma - delta_pa * pa

    dmb_dt = +gamma_mb - delta_mb * mb
    dpb_dt = +gamma_pb * mb - delta_pb * pb

    dmc_dt = +gamma_mc - delta_mc * mc
    dpc_dt = +gamma_pc * mc - delta_pc * pc

    return [dma_dt, dpa_dt, dmb_dt, dpb_dt, dmc_dt, dpc_dt]

sol = solve_ivp(dxdt, [0.0, 400], [0, 0, 0, 0, 0, 0])

T = sol.t
X = sol.y.T

fig, (ax_a, ax_b, ax_c) = plt.subplots(3, 1, figsize=(6, 7))

for ax, (i_m, i_p), name in zip((ax_a, ax_b, ax_c), ((0, 1), (2, 3), (4, 5)), "ABC"):
    ax.plot(T, X[:, i_m], "-r") # red for RNA
    ax.plot(T, X[:, i_p], "--", color=[0.5, 0, 1]) # purple
    ax.set_ylabel(f"RNA and Protein {name}")
    ax.set_xlabel("Time (seconds)")

fig.tight_layout()

```

```
plt.show()
```

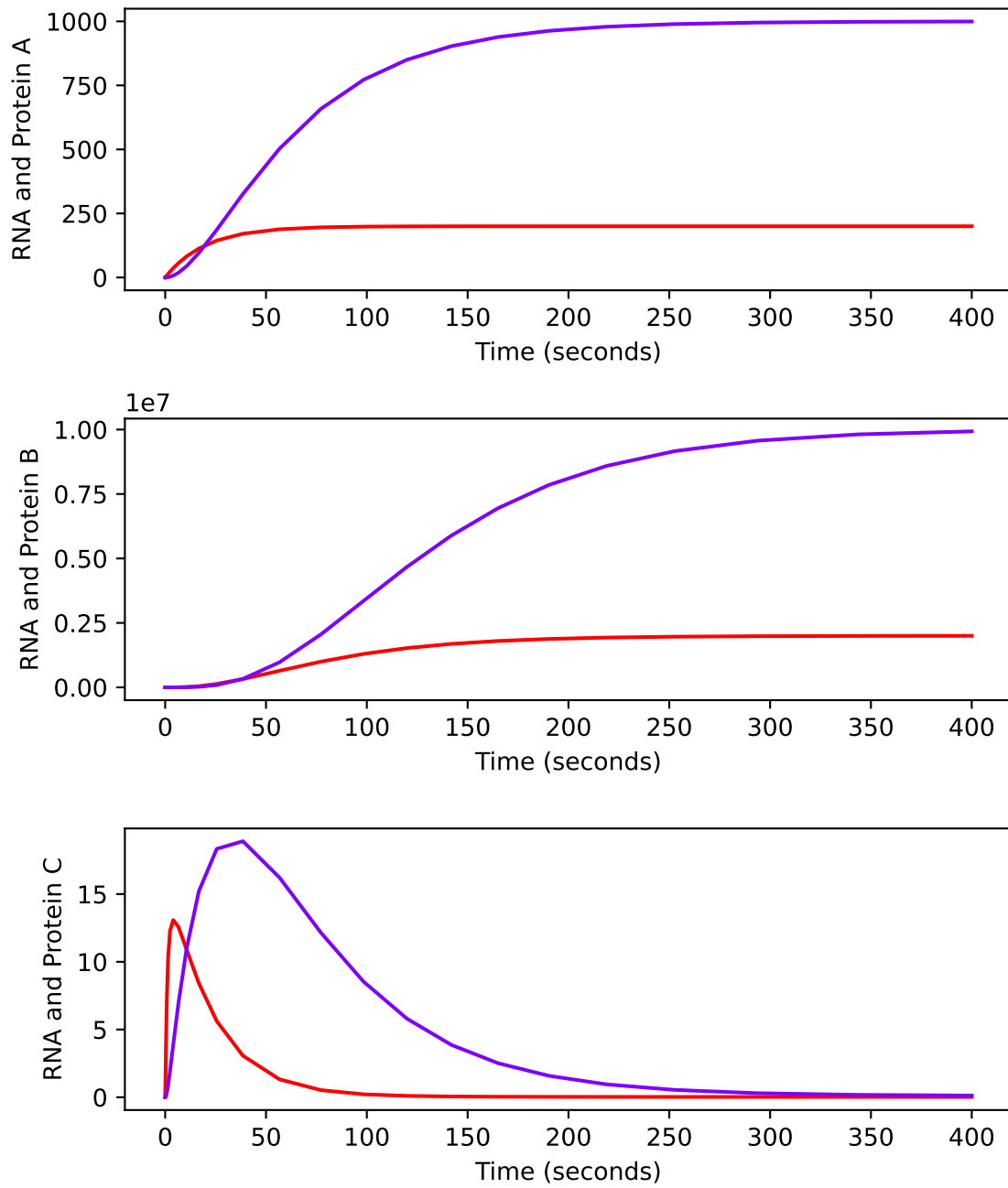


Figure 6: Incoherent feed-forward. The pulse in the bottom panel is the motif's signature.

5. The Repressilator

```
delta_ma = 100 * 0.05
gamma_pa = 100 * 0.1
delta_pa = 100 * 0.02

delta_mb = 0.05
gamma_pb = 0.1
delta_pb = 0.02

delta_mc = 0.05
gamma_pc = 0.1
delta_pc = 0.02

Kca = 100 # Strength (IC50) of inhibition of A by C
Kab = 100 # Strength (IC50) of inhibition of B by A
Kbc = 100 # Strength (IC50) of inhibition of C by B

def dxdt(t, state):
    """A ring of three repressors: C represses A, A represses B, B represses C."""
    ma, pa, mb, pb, mc, pc = state

    gamma_ma = 10 / (1 + (Kca * pc))
    gamma_mb = 10 / (1 + (Kab * pa))
    gamma_mc = 10 / (1 + (Kbc * pb))

    dma_dt = +gamma_ma - delta_ma * ma
    dpa_dt = +gamma_pa * ma - delta_pa * pa

    dmb_dt = +gamma_mb - delta_mb * mb
    dpb_dt = +gamma_pb * mb - delta_pb * pb

    dmc_dt = +gamma_mc - delta_mc * mc
    dpc_dt = +gamma_pc * mc - delta_pc * pc

    return [dma_dt, dpa_dt, dmb_dt, dpb_dt, dmc_dt, dpc_dt]

sol = solve_ivp(dxdt, [0, 1e5], [0, 0, 0, 0, 0, 0])

T = sol.t
```

```

X = sol.y.T

fig, (ax_a, ax_b, ax_c) = plt.subplots(3, 1, figsize=(6, 7))

for ax, (i_m, i_p), name in zip((ax_a, ax_b, ax_c), ((0, 1), (2, 3), (4, 5)), "ABC"):
    ax.plot(T, X[:, i_m], "-r") # red for RNA
    ax.plot(T, X[:, i_p], "-", color=[0.5, 0, 1]) # purple
    ax.set_ylabel(f"RNA and Protein {name}")
    ax.set_xlabel("Time (seconds)")
    ax.set_xscale("log")
    ax.set_xlim(1e-4, 1e5)

fig.tight_layout()
plt.show()

```

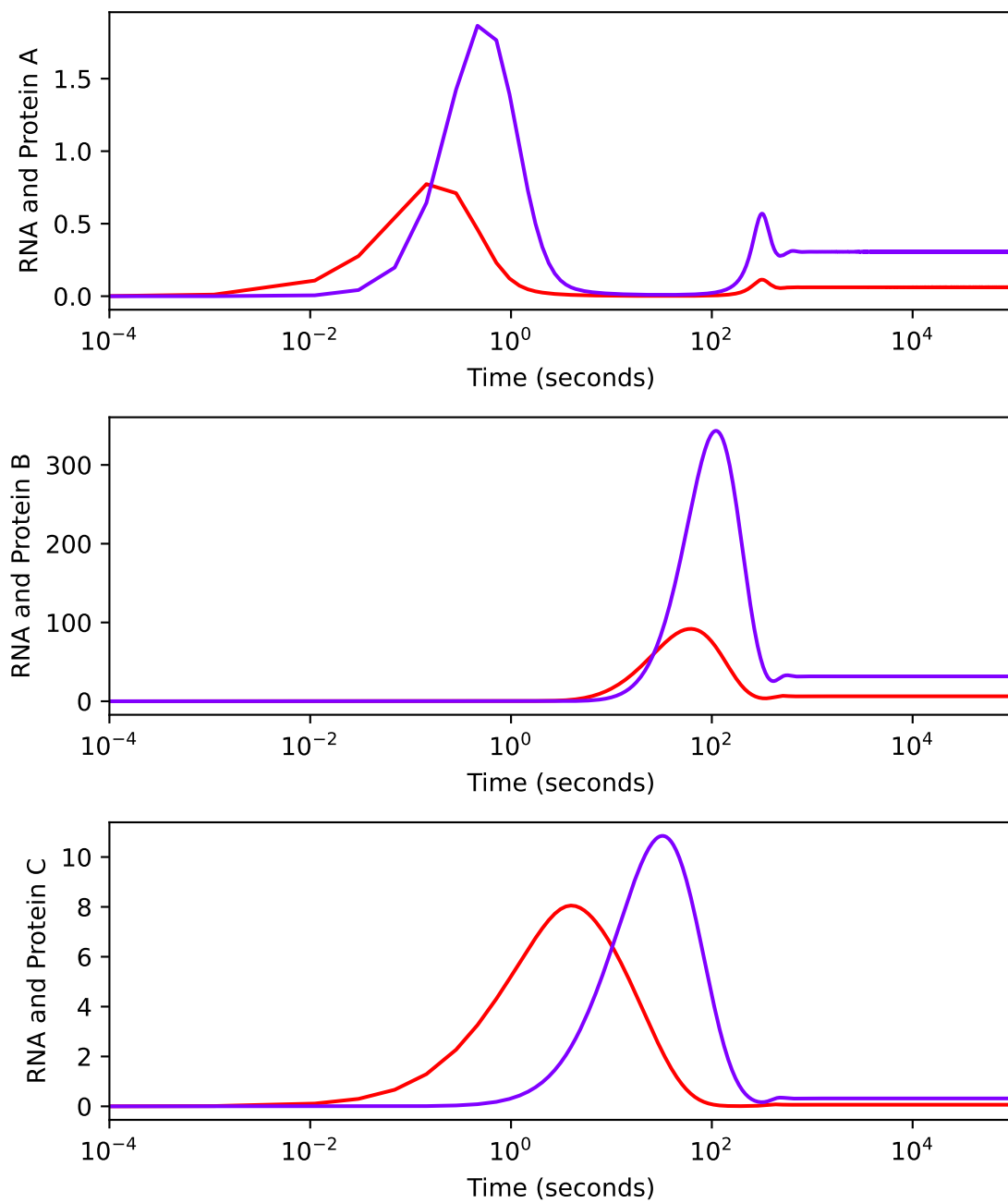


Figure 7: The Repressilator, on a logarithmic time axis so that all three genes are visible at once.